

VehicleIQ: Real-Time Vehicle Health & Failure Prediction System

B.Tech CS (AI/ML) Final Year Technical Project Report | Author: Jyotiraditya Patil

1. Executive Summary & Abstract

VehicleIQ introduces a proactive, end-to-end AI framework that ingests continuous real-time OBD-II sensor streams (RPM, Speed, Coolant Temp, Throttle, Fuel Trim, Battery Voltage, MAF) and applies predictive machine learning models to forecast component degradation before failure occurs. The platform includes sliding-window feature extraction, a Random Forest Driving Style Classifier (Gentle, Moderate, Aggressive profiles), an LSTM Neural Network predicting multi-component health scores (Brake Wear, Engine Stress, Battery Health), and a SHAP Explainability Engine outputting plain-English diagnostic alerts and step-by-step DIY repair solutions.

2. Technology Stack & Tools Used

Category	Technology	Project Usage
Languages	Python 3.11, TypeScript	Backend, ML Models, Next.js Web App
Ingestion	python-OBD, Pandas, NumPy	Serial PID reading, noise filtering, cleaning
Machine Learning	Scikit-Learn, TensorFlow, SHAP	Random Forest classifier, LSTM predictor, SHAP alerts
Backend & Security	FastAPI, SQLAlchemy, PyJWT	Async REST API, ORM DB queries, JWT Security
Web Dashboard	Next.js 14, React, Glassmorphic CSS	Live instruments, component health rings, fix guides
DevOps & Cloud	Docker, Render, Vercel, GitHub Actions	Multi-container setup, cloud deployment, CI/CD

3. Implementation Methodology & Phase Breakdown

Phase 1: OBD-II Bluetooth & physics simulator ingestion engine with Pandas cleaning and SQLAlchemy DB persistence.

Phase 2: Behavioral feature extraction (hard braking rate, high RPM ratio, thermal stress) with Random Forest & LSTM failure models.

Phase 3: FastAPI REST API backend with JWT auth, Next.js 14 web dashboard with live instruments, and printable mechanic reports.

Phase 4: React Native Expo mobile app, Render/Vercel deployment configs, and GitHub Actions CI/CD pipeline.

4. Future Scope & Improvements

- CAN-Bus Deep Decoding:** Parsing vendor-specific ISO 15765-4 CAN frames for tire pressure and transmission temp.
- Edge Microcontroller AI:** Quantizing LSTM models with TensorFlow Lite for Raspberry Pi / Jetson Nano in-vehicle inferencing.
- Federated Fleet Learning:** Collaborative multi-vehicle failure prediction without centralized log transmission.
- Automated Parts Booking:** AutoZone / NAPA API integration for 1-click replacement part ordering.